

Language	API/Binding
C/C++	Provided by SQLite development team
Java	SQLite-JDBC (Most popular) https://bitbucket.org/xerial/sqlite-jdbc/overview
PHP	PECL extension
Python, Perl	Available as part of language libraries
Ruby	<i>sqlite3-ruby</i> as a gem
JavaScript	JSPDO offers a driver/API for SQLite

For Android, Google has embedded the database engine in the OS itself, and has a Java API for applications to use. Apple also adopted it for iOS and made it a part of the core Data API in both iOS as well as in Mac OS X 10.4, which enables developers to use it in both iOS as well as desktop apps. In the case of desktop Web apps, Firefox uses SQLite widely for storing bookmarks and cookies. Chrome and Opera use it for storing internal configuration and WebSQL databases. Skype and Adobe also use it for storing internal data. There is a Firefox add-on, SQLite Manager, to view, create, edit, and update SQLite databases, which is available from the Firefox add-ons site. And an Android application to view, create and edit SQLite databases is also available in the Play store.

Using SQLite in lightweight PHP Web apps

SQLite is part of PHP releases from 5.0 till 5.3. From version 5.4 onwards, SQLite is available as a PECL extension. SQLite and PDO drivers must be installed and enabled (*extension=php_pdo.so, extension=php_sqlite.so in /etc/php.ini*). To check the status, use the following code:

```
$ echo `<?php phpinfo() ?>` > test.php
$ php test.php | grep -i sqlite
PDO Driver for SQLite 3.x => enabled
SQLite Library => 3.6.29
sqlite3
SQLite3 support => enabled
```

Step 1: Create a database—either as a file or in-memory. If created in a file, ensure that the *apache/www* user has RW access to the file.

```
//any name other than 'memory:' is identified
//as a database in a file.
$in_mem_db = new PDO('sqlite::memory:');
```

Step 2: Create tables in the DB to track OSFY visitors and the number of their visits:

```
$in_mem_db->exec("CREATE TABLE IF NOT EXISTS
osfyReaders(
    id INTEGER PRIMARY KEY,
    name TEXT,
    visits INTEGER)");
```

Step 3: Create an array containing a list of users and bind the variables to the columns of the table, and fill in the details:

```
// prepare insert statement
$insert = "INSERT INTO osfyReaders (name, visits)
VALUES (:name, :visits)";

$stmt = $in_mem_db->prepare($insert);

// Bind parameters to statement variables
$stmt->bindParam(':name', $name);
$stmt->bindParam(':visits', $visits);

// Loop through all visitors and execute prepared
insert statement
foreach ($readers as $r) {
    // Set values to bound variables
    $name = $r['name'];
    $visits = $r['visits'];

    // Execute statement
    $stmt->execute();
}
```

Step 4: Now retrieve the data from the table with a query, and display it:

```
// Select all data from in memory db
osfyReaders table
$result = $in_mem_db->query('SELECT * FROM
osfyReaders');

foreach($result as $row) {
    echo "Id: " . $row['id'] . "\n";
    echo "Name: " . $row['name'] . "\n";
    echo "Visits: " . $row['visits'] . "\n";
    echo "\n";
}
```

Step 5: Close the connection to the database.

```
$in_mem_db = null;
```

The SQLite package also provides a command-line tool, *sqlite3*, which can be used to explore all the capabilities of the database. To list all available options, enter *.help* at the prompt.